

A Late-Time Propagation Correction as a Minimal Resolution to the H_0 Tension

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A Late-Time Propagation Correction as a Minimal Resolution to the H_0
Tension
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Abstract

The discrepancy between early-universe and late-universe measurements of the Hubble constant remains one of the most significant open problems in cosmology. Measurements from the Planck Collaboration yield $H_0 \approx 67.4$ km/s/Mpc, while local distance ladder measurements report $H_0 \approx 73$ km/s/Mpc.

We propose a minimal phenomenological extension to Λ CDM in which a small late-time propagation correction modifies luminosity distances without altering early-universe physics. The model introduces a redshift-dependent propagation parameter $\delta(z)$ that vanishes at high redshift and reaches a finite value at $z \approx 0$. We demonstrate that a present-day correction $\delta_0 \approx 0.077$ reconciles the observed discrepancy while preserving Cosmic Microwave Background constraints.

1 Introduction

The Hubble constant tension refers to the statistically significant discrepancy between early-universe inference from CMB anisotropies and late-universe direct measurements using Type Ia supernovae.

Planck 2018 results give:

$$H_0^{\text{CMB}} \approx 67.4 \text{ km/s/Mpc}, \quad (1)$$

while local distance ladder measurements give:

$$H_0^{\text{local}} \approx 73 \text{ km/s/Mpc}. \quad (2)$$

The discrepancy exceeds 5σ significance.

2 Model Framework

We introduce an effective propagation velocity:

$$V_{\Omega}(z) = \frac{C}{1 + \delta(z)}, \quad (3)$$

where C is the standard speed of light and $\delta(z)$ is a dimensionless propagation correction.

To preserve early-universe observables:

$$\delta(z \gg 1) \rightarrow 0. \quad (4)$$

3 Parametrization

We adopt:

$$\delta(z) = \delta_0(1 + z)^\alpha, \quad (5)$$

with

$$\alpha < -2. \quad (6)$$

To reconcile the Hubble tension:

$$\frac{H_0^{\text{obs}}}{H_0^{\text{true}}} = \frac{1}{1 - \delta_0}. \quad (7)$$

Using observed values:

$$\frac{73}{67.4} \approx 1.083, \quad (8)$$

we obtain:

$$\delta_0 \approx 0.077. \quad (9)$$

4 Luminosity Distance Modification

In Λ CDM:

$$D_L(z) = (1 + z) \int_0^z \frac{C}{H(z')} dz'. \quad (10)$$

In the present model:

$$D_L^\Omega(z) = (1 + z) \int_0^z \frac{C}{(1 + \delta(z'))H(z')} dz'. \quad (11)$$

For $z \ll 1$:

$$D_L^\Omega \approx \frac{C}{H_0} (1 - \delta_0) z. \quad (12)$$

Thus:

$$H_0^{\text{obs}} = \frac{H_0^{\text{true}}}{1 - \delta_0}. \quad (13)$$

5 Consistency with CMB

The angular acoustic scale:

$$\ell_A = \pi \frac{D_A(z_*)}{r_s(z_*)}. \quad (14)$$

Since:

$$\delta(z_*) \approx 0 \quad for \alpha < -3, \quad (15)$$

CMB constraints remain intact.

6 Observational Tests

The model predicts:

- No modification to CMB peak structure
- No alteration of Big Bang nucleosynthesis
- A systematic shift in low-redshift distance ladder measurements
- Potential deviations in low- z BAO observations

Future tests include:

- Pantheon supernova χ^2 fitting
- Joint BAO + SN analysis
- Gravitational wave standard siren constraints

7 Discussion

This framework does not eliminate dark energy. It introduces a late-time propagation correction that modifies inferred expansion rates without altering early-universe cosmology.

Challenges include:

- Laboratory constraints on light propagation
- Gravitational wave speed constraints
- Solar system Shapiro delay limits

8 Conclusion

We present a minimal late-time propagation correction capable of reconciling the Hubble tension while preserving Λ CDM consistency at high redshift.

The required present-day correction is:

$$\delta_0 \approx 0.077. \quad (16)$$

Further work must evaluate compatibility with precision astrophysical and laboratory constraints.

Acknowledgments

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